



Dune Ops Observability

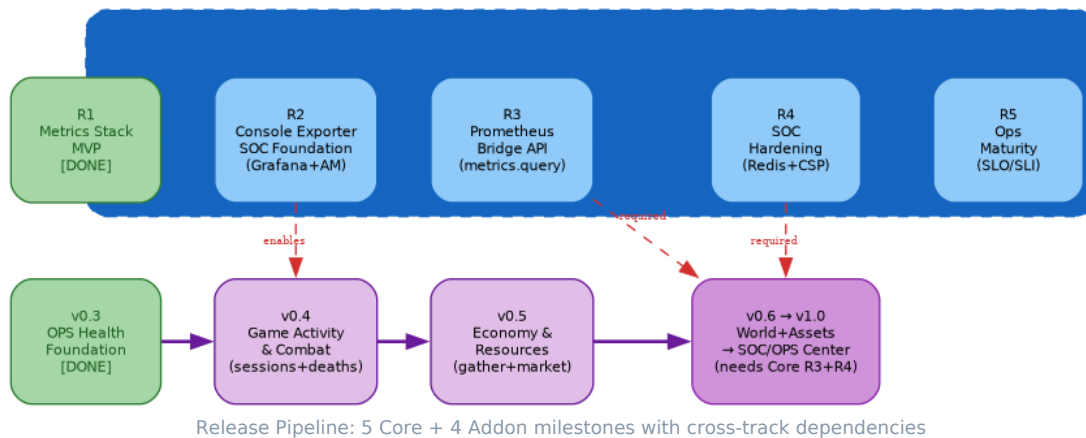
Comprehensive Roadmap

Core R1-R5 | Addon v0.2-v1.0 | July 2026
yacketrj/dune-ops-observability-addon

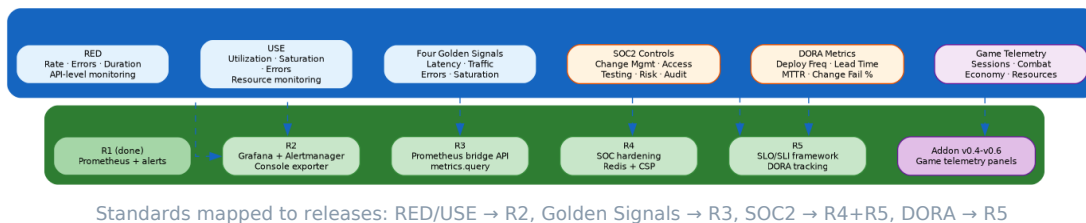
Executive Summary

Dune Ops Observability is a community addon providing SOC/OPS monitoring, game telemetry analytics, and formal release governance for Red-Blink's Dune: Awakening self-hosted Docker server. The Core repository provides infrastructure (Prometheus, Grafana, Alertmanager, bridge API). The Addon provides operator-facing panels rendered inside the Console iframe.

This roadmap defines 5 Core releases and 6 Addon releases with explicit cross-track dependencies, 8 tag types, and mapping to industry-standard metrics (RED, USE, Golden Signals, SOC2, DORA).

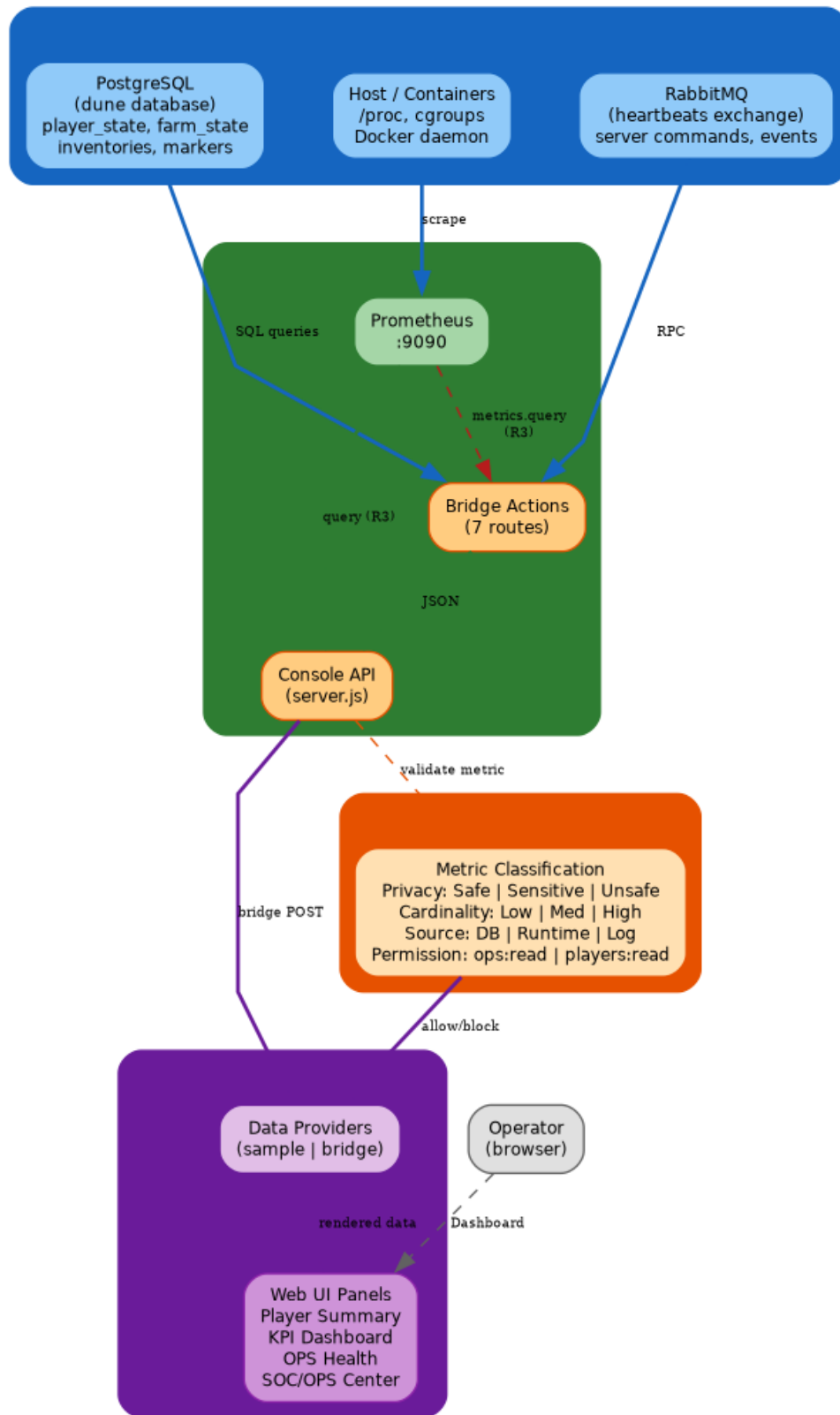


Industry Standards Mapping



Core Release Train

R	Name	Key Deliverables	Tags
R1	Metrics Stack MVP	Prometheus + 4 exporters, 16 alerts, CLI, SOC controls	v1.3.x (upstream)
R2	Console Exporter + SOC Found	Grafana (always-on), Alertmanager (email+webhook), game metrics	core-metrics-r2
R3	Prometheus Bridge API	metrics.query bridge, /api/server/metrics, structured logging	core-metrics-r3
R4	SOC Hardening	Redis rate limits, proxy-aware IP, per-addon CSP	core-soc-r4
R5	Ops Maturity	SLO/SLI framework, retention/backup, audit rotation, DORA	core-ops-r5

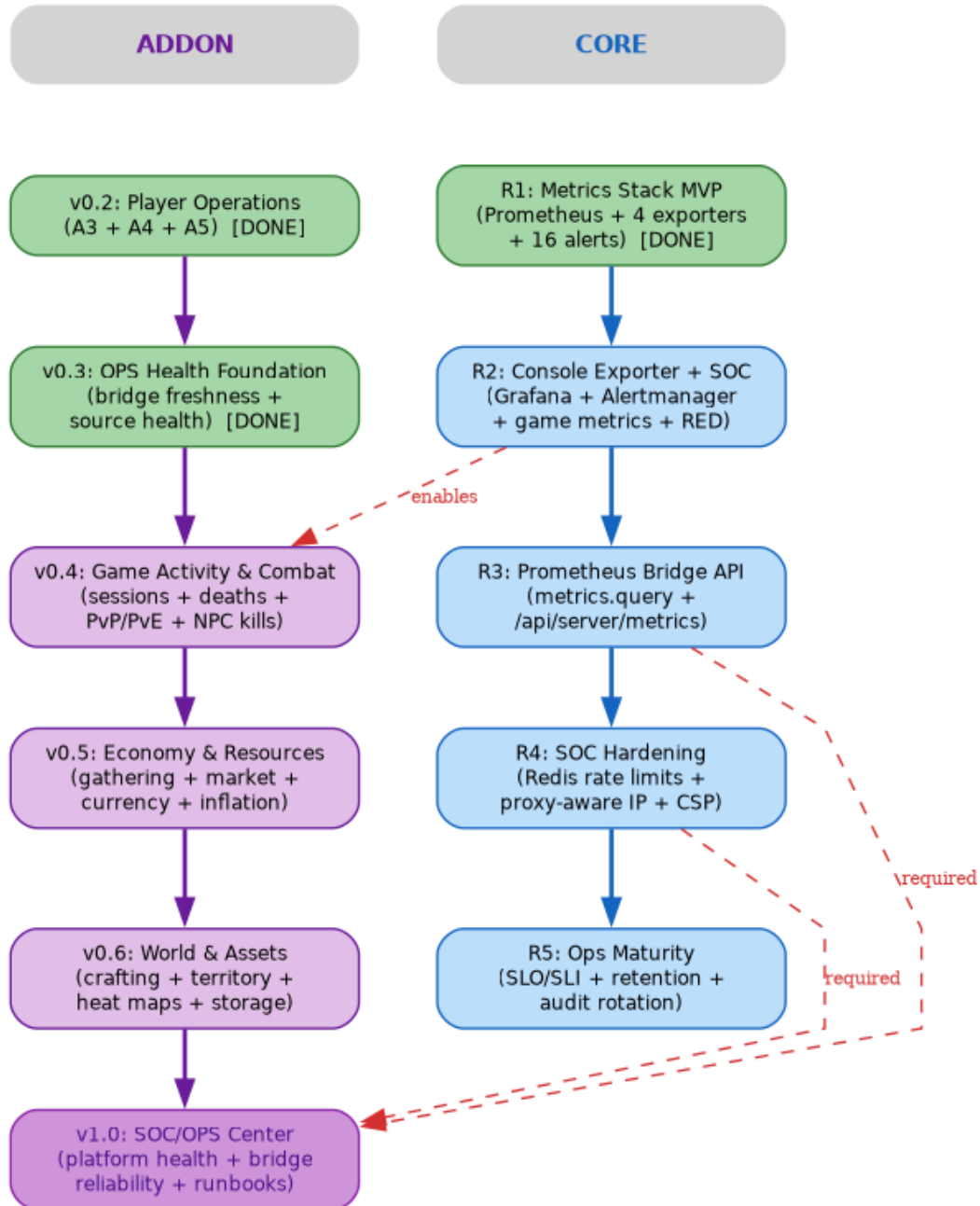


Data flow: Game DB → Core Bridge → Addon Panels → Operator Dashboard

Addon Release Train

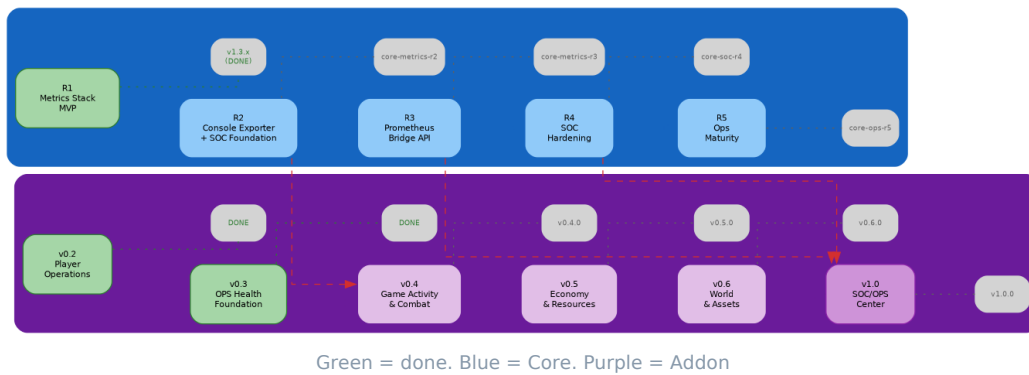
Release	Name	Merges	Key Metrics	Dep	Tags
v0.2.0	Player Operations	A3+A4+A5	Player summary, KPI, capability	-	v0.2.0
v0.3.0	OPS Health Foundation	released	Bridge freshness, source health, operator s	-	v0.3.0
v0.4.0	Game Activity & Comb	v0.4+v0.5	Sessions, deaths, PvP/PvE, NPC kills	R2	v0.4.0
v0.5.0	Economy & Resources	v0.6+v0.7	Gathering, currency, market, inflation	-	v0.5.0
v0.6.0	World & Assets	v0.8+v0.9	Crafting, territory, heat maps, storage	-	v0.6.0
v1.0.0	SOC/OPS Center	full platform	Platform health, bridge reliability, Prometh	R3+R4	v1.0.0

Dependency Graph

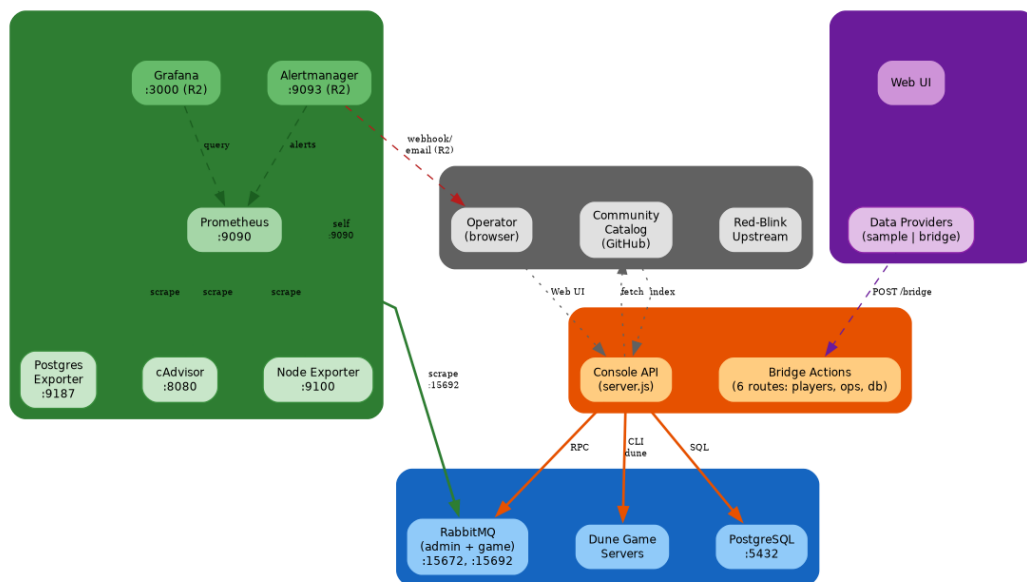


Solid: Core pipeline + Addon pipeline. Red dashed: Core-to-Addon coupling

Release Timeline



Architecture



Game Stack, Metrics Stack, Console API, Addon, External

Database Discovery Phase

Before any game-facing metric in v0.4-v0.6, a PostgreSQL event inventory must run against dune-postgres. This is a mandatory gate.

Required Output

- Schemas, tables, views, columns with types
- Candidate event/audit/activity tables + timestamp coverage + retention
- Estimated row counts
- Metric-source mapping: name → schema.table → columns → aggregation → cardinality → privacy

Discovery Findings (from first run)

- Available: player state, farm state, inventory (79 items), resource fields (27 entries), markers (41)
- Blocked (zero rows): game_events, event_log, exchange_orders, fulfilled_orders
- Approved safe: player status summary, farm aggregate (total/ready/alive/S2S)

Hard rule: No economy, combat, death, NPC, resource, crafting, inventory, market, or location metric enters a release without confirmed source.



Discovery process: Inventory → Classify → Approve/Block → Implement

Tagging Convention

Context	Pattern	Example	Repo
Core feature merges	core-{domain}-r{n}	core-metrics-r2	Core fork
Addon releases	v{m}.{m}.{p}	v0.4.0, v1.0.0	Addon
Addon RCs	v{m}.{m}.{p}-rc{n}	v0.4.0-rc1	Addon
Addon features	feature/{area}	feature/player-activity	Addon
Evidence archive	evidence/v{m}.{m}.{p}	evidence/v0.4.0	Addon
Immutable snapshots	preserve/{desc}	preserve/pre-db-discovery	Addon
Core feature branches	feature/{area}	feature/console-exporter	Core fork

Release Cadence

- Addon PRs: focused, frequent. Public releases: $\leq$ every 2 weeks
- Core releases: batched by capability block (R1→R5)
- Upstream PRs: after fork validation, narrow scope, evidence-backed
- Emergency patches: allowed for broken pkg, invalid manifest, security

Versioning (semver): PATCH=bug/package/manifest/UI fix. MINOR=new panel/feature/workflow.
MAJOR=permission expansion/breaking manifest/incompatible bridge.

SOC/OPS Priority Matrix

P0 — Critical

Area	Metrics	Core
Addon & Bridge Health	bridge available, errors, last read, data age	-
Players	online/offline, active rate, last seen staleness	-
Console/API	request errors, latency, timeouts	R2 (RED)
Capacity & Saturation	CPU, memory, disk, DB size, container health	R2 (Grafana)
Service Lifecycle	up/down, restart count, degraded	R2 (Alertmanager)
Grafana + Alertmanager	up/down, dashboard health, alert routing	R2
Metrics Bridge	metrics.query availability, rejection rate	R3

P1 — Security-Sensitive

Area	Metrics	Core
Admin Activity	commands, privilege changes, config changes	-
Access Failures	failed auth, denied addon requests, permission failures	R4 (Redis)
Addon Integrity	manifest drift, checksum drift, permission drift	-
Abuse Signals	suspicious churn, ban/kick trends	R4 (proxy IP)
Rate Limit Persistence	bucket state, abuse trends, IP rotation detection	R4 (Redis)
Per-addon CSP	CSP violations, blocked outbound requests	R4

P2 — Operational Analytics

Area	Metrics	Core
Faction & Guild	faction/guild share, top faction/guild	-
Economy	spice, solari, market, exchange, tax	DB discovery
API RED Metrics	request rate, error rate, p50/p95/p99	R2 (exporter)
Bridge Analytics	call rate, error rate, response time	R3
SLO/SLI	availability %, latency SLO, freshness SLO	R5

Metric Classification

Privacy Class	Definition	Examples	Action
Safe	May expose after review	player counts, farm counts, container status	Allowed
Sensitive	Needs additional review	timestamps, log-derived counts, host paths	Review
Unsafe	Must NOT expose	player IDs, names, coords, tokens, secrets	Blocked
Unknown	Classify first	opaque blobs, unclear enums	Hold

Security Boundaries

Layer 1: Transport
HMAC-SHA256 sessions
CSRF tokens on mutations
Security headers (nosniff, DENY)
timingSafeEqual password compare

Layer 2: Rate Limiting
Login: per-client 8/15m + global 32/15m
Bridge: per-addon 60/min + global 300/min
R4: Redis-backed persistence + proxy-aware IP

Layer 3: Input Validation
JSON body size limit (2MiB)
Upload size limit (1GiB)
IP allowlisting (ADMIN_ALLOWED_IPS)
Addon ID pattern validation (`^[a-z0-9-]{1,63}$`)

Layer 4: Bridge Access Control
Permission gating (ops:read, players:read)
Read-only SQL enforcement
SHA-256 checksum verification
Addon provenance tracking

Layer 5: Data Protection
Secret redaction (JWT, tokens, passwords)
JSONL audit logging
R4: Per-addon CSP sandbox
R4: Proxy-aware IP detection

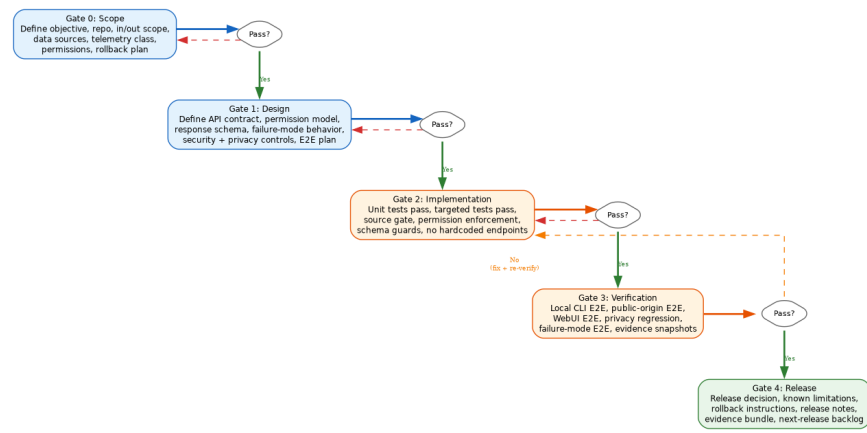
Attack Surface (R2-R5)

Component	Current	R2-R5	Mitigation
Prometheus	127.0.0.1:9090	unchanged	Localhost-only
Grafana (R2)	n/a	127.0.0.1:3000	Auto-gen admin password
Alertmanager (R2)	n/a	127.0.0.1:9093	Webhooks over TLS
Redis (R4)	n/a	dune-net:6379	Internal net, opt AUTH
metrics.query (R3)	n/a	POST bridge	ops:read, PromQL sanitize

Non-Negotiable Security Rules

- raw player rows, player IDs, account IDs, character names
- Funcom/FLS identifiers, actor IDs, coordinates, exact positions
- SQL text, PromQL text, tokens, passwords, secrets
- raw logs by default, unbounded high-cardinality labels

Release Process: 5 Gates



Scope → Design → Implementation → Verification → Release

Gate	Name	Exit Criteria
0	Scope	Objective, repo, in/out scope, data sources, permissions, rollback plan
1	Design	API contract, permission model, response schema, failure-mode, E2E plan
2	Implementation	Unit + targeted tests, permission enforcement, schema guards
3	Verification	Local CLI E2E, WebUI E2E, privacy regression, evidence snapshots
4	Release	Decision, limitations, rollback, release notes, 16-file evidence bundle

Governance & Definition of Done

Feature Branch
→ PR opened

Clean main
Lockstep with origin

Unit tests
pass

Security scans
(Gitleaks, Semgrep, Trivy)

Documentation
impact reviewed

PR Gates (minimum)

- main was clean and in lockstep with origin/main before branching
- PR body: Summary, Why, Documentation Impact, Testing, Security, Risks
- Gitleaks, Semgrep, Trivy output included or justified
- SBOM impact stated; SOC2-style controls documented

Release Gates

- Test + Security evidence captured (unit, regression, E2E, Gitleaks, Semgrep, Trivy)
- 16-file evidence bundle, SBOM, SOC2-style controls
- Release decision: Approved | Approved w/limits | Blocked | Abandoned | Superseded
- Package built, SHA-256 verified, GitHub Release published, catalog PR submitted

Decision Log

Decision	Date	Rationale
v0.4-0.6 grouped into 3 releases	2026-07-04	v0.4=activity+combat, v0.5=economy+resources, v0.6=world+assets
Grafana always-on	2026-07-04	Operators need persistent access. Localhost by default
Alertmanager email+webhook	2026-07-04	Email basic, webhook (Discord/Slack) for team ops
Redis for rate limits	2026-07-04	Small dep, survives restart. Internal net only
v1.0 waits for R3+R4	2026-07-04	Needs metrics.query (R3) + persistent rate limits (R4)
Addon-first architecture	2026-06-30	Addon=UI, Core=infrastructure per RFC Addendum
DB discovery mandatory	2026-07-02	No metric until source confirmed via inventory
16-file evidence bundle	2026-06-30	Per release-standard. SOC2-style governance
Command-auth upstream-compatible	2026-07-04	Built-in fallback required. Game server uses same token

Document Index

Roadmap & Planning

[ROADMAP.md](#)
[OBSERVABILITY-ROADMAP.md](#)
[SOC-OPS-ROADMAP.md](#)
[RFC.md](#)

- unified overview, dependency graph, tagging
- per-release candidate metrics, DB review
- P0/P1/P2 taxonomy, Core R2-R5
- formal RFC

Governance & Standards

[release-standard.md](#)
[metric-classification-standard.md](#)
[REPOSITORY-REQUIREMENTS-AND-DELIVERABLES.md](#)

- 5-gate process, evidence bundle, security rules
- privacy classes, cardinality, DB rules
- PR/release gates, SBOM, SOC2, DoD

Implementation Evidence (Core)

[R1-METRICS-STACK-IMPLEMENTATION-NOTES.md](#)
[PR-EVIDENCE-ADDON-METRICS-SUPPORT.md](#)
[E2E-METRICS-TESTING.md](#)

- R1 design, security, validation
- metrics stack PR scope, E2E
- E2E test procedure

Last updated: 04 July 2026. Maintained in [yacketrj/dune-ops-observability-addon](#).